

Modeling the Emergence of Indexical Order through Local Interaction

GitHub Repo: https://github.com/juliannayufan/ABM_Projects/tree/main/Final_Project

External Resources:

- Mesa documentation for API reference and implementation guidance.
- ChatGPT and Copilot were used for debugging and for clarifying some Mesa concepts.

1. Introduction and Research Question

How do shared interpretations of an object or topic emerge and stabilize within social interaction? Traditional approaches in linguistic anthropology have emphasized that meaning is not fixed but socially produced through interaction. However, less attention has been paid to how local interactional dynamics scale up into stable or unstable macro-level interpretive structures. This ABM project intends to simulate and address this gap by asking: *How do indexical interpretations of a shared object/topic emerge, stabilize, or fail to stabilize within a social network through repeated local interaction?*

The concept of indexicality provides a useful entry point to this problem. In linguistic anthropology, indexicality refers to the process through which linguistic forms become socially meaningful not only through reference but through their ability to point to social contexts, identities, and stances (Silverstein 2003; Agha 2007). Rather than being fixed semantic units, meanings are dynamically produced in interaction. Building on this insight, scholars such as Agha (2003) and Blommaert (2015) have argued that indexical meanings become socially organized through processes of enregisterment, scaling, and interactional circulation. These processes explain how certain ways of speaking become recognizable as socially meaningful “registers,” while others remain locally restricted or fail to stabilize across contexts. However, while rich ethnographic and theoretical work has demonstrated how indexical meanings are socially constructed, less is known about how such processes scale over repeated interactions

in structured populations. In particular, the relationship between micro-level interaction and the emergence (or non-emergence) of stable indexical structures remains under-theorized in computational terms.

This study uses ABM modeling to make these otherwise abstract theoretical processes visible and experimentally manipulable. Rather than treating indexical meaning as either fully stable or fully fluid, the model treats it as a dynamic and bounded interpretive space in which agents continuously adjust their positions through interaction. Importantly, the model does not assume that consensus is the natural or inevitable outcome of interaction. Instead, it explicitly allows for divergence, partial stabilization, and non-convergence, in line with critiques of over-deterministic models of agreement formation in social systems.

2. Literature Review (Theoretical Premises)

Indexicality refers to the way linguistic and semiotic forms come to carry meanings that go beyond direct reference, acquiring social significance through use (Silverstein, 2003; Agha, 2007). Meaning in this sense is not fixed in advance or contained in forms themselves, but emerges through repeated contextual encounters in which particular expressions become associated with recognizable social interpretations. Interpretation is therefore fundamentally historical and interactional: what a form “means” depends on how it has been used and taken up across situations, which motivates modeling agents not as carriers of stable meanings but as actors whose interpretations shift through interaction. Closely related to this is the idea that some patterns of meaning can become socially stabilized when they are repeatedly recognized and linked to social types, situations, or values. Through this process, certain ways of using language or interpreting signs become conventionalized and begin to function as shared interpretive resources. However, this stabilization is never guaranteed. Many interpretations remain local, contested, or short-lived, and even widely circulating meanings can fail to

stabilize across different contexts. This makes interpretive stability an empirical outcome rather than an assumption, directly informing the model's focus on when alignment emerges and when it does not.

A key condition shaping this process is that meaning always operates across different scales of circulation and recognizability (Blommaert, 2015). An interpretation may be highly intelligible within a small interactional setting but lose force or relevance when extended to broader social contexts. Conversely, some meanings become widely recognizable and travel across contexts with relative stability. This idea is operationalized in the model through the parameter K , which defines the resolution of the interpretive space in which agents operate. Rather than representing correctness or ability, K controls how finely interpretations can be differentiated. Lower values compress the space and make convergence easier, while higher values increase interpretive diversity and make stable alignment more difficult to achieve.

In addition to indexicality and scales, meaning is also produced through interaction itself, rather than existing prior to it. Interactional dynamics are grounded in Goffman's (1967) notion of interaction ritual, where meaning is jointly produced through situated encounters rather than individually held representations. In face-to-face interaction, participants continuously adjust their interpretations based on others' responses, and meaning emerges through these local coordination processes. In the model, this is operationalized by placing agents in a spatial network where they only interact with their immediate neighbors. Two interaction structures are distinguished. In dyadic interaction, meaning adjustment occurs through one-to-one encounters, producing relatively direct alignment pressure. In group interaction, multiple interaction partners are involved at once, introducing competing interpretive influences that make convergence more variable and less predictable.

The temporal organization of interaction further shapes these dynamics through different forms of turn-taking and coordination. Interaction is not simply simultaneous exchange but is structured through sequential turns, where each participant's action both responds to and reshapes the conditions for the next move (Sacks et al. 1974). Drawing on the basic insight that interaction is sequentially organized and timing-sensitive, interaction can unfold (1) in a fully synchronized way, where updates are coordinated across the whole system, (2) in an asynchronous way, where agents update sequentially and influence one another in real time, or (3) in a partially synchronized way, where updates occur in batches. These different temporal regimes matter because they affect how quickly and how consistently alignment can spread through the system.

Across all mechanisms applied in the model, alignment is probabilistic rather than deterministic, ensuring that consensus is not structurally guaranteed. Stabilization is therefore treated as an emergent outcome produced by the three mechanisms building off of the above theories: (1) interaction structure, (2) scalar resolution (K), and (3) temporal update regime. This allows the model to examine when indexical interpretations stabilize, remain plural, or fail to converge under different social configurations.

3. Model Description

The model represents a population of 200 agents positioned on a two-dimensional grid with single occupancy. Each agent represents a situated interpretive actor engaged in making sense of a shared object. Each agent is initially assigned a scalar position within a bounded discrete space $[0, K]$, where K defines the resolution of possible interpretive distinctions. In this sense, the agent's state represents its real-time interpretations rather than competence or truth, so that each agent's interpretive position is continuously updated through interaction with local neighbors under different structural and temporal conditions.

The model implements two distinct interaction structures. In **dyadic** interaction, an agent engages with a single randomly selected neighbor at each update step. The agent compares its own interpretive position with that of its partner and adjusts by moving one step closer along the interpretive scale. This produces incremental alignment driven by pairwise comparison. In **group** interaction, an agent instead responds to multiple neighbors simultaneously by computing the average state of its local neighborhood. The agent then moves one step toward this local mean, producing a weaker but more distributed alignment pressure. Both interaction types are probabilistic and do not guarantee change, reflecting the fact that interpretive adjustment in real interaction is contingent rather than automatic.

In addition to interaction structure, the model incorporates a temporal mechanism governing how updates are scheduled. Three timing regimes are implemented: synchronous, asynchronous, and partial synchrony. In **synchronous** updating, all agents first determine their intended updates before any changes are applied, producing a globally coordinated update cycle. In **asynchronous** updating, agents update sequentially in randomized order, meaning that earlier interactions can immediately influence later ones, generating path-dependent dynamics. **Partial** synchrony divides agents into batches, where updates occur within local groups before moving to the next batch, producing intermediate coordination between global and fully sequential update regimes. These temporal structures operationalize different forms of interactional coordination and turn-taking organization, affecting how interpretive influence propagates through the system.

Moreover, two additional probabilistic mechanisms regulate whether interaction leads to actual change. First, **alignment probability** controls whether an interaction results in any movement along the interpretive scale. Second, **inertia** introduces resistance to change, meaning that even when interaction occurs, agents may retain their current interpretive position. These

mechanisms ensure that convergence is not mechanically enforced but emerges only when repeated interactional pressure overcomes stochastic resistance.

Overall, the model operationalizes indexical interpretation as a dynamic process emerging from the interaction of three mechanisms: (1) interaction structure (dyadic vs group), which determines the local configuration of interpretive influence; (2) scalar resolution (K), which defines the granularity of possible interpretive distinctions; and (3) temporal update regime (synchronous, asynchronous, partial), which governs how interactional influence is sequenced over time. Together, these mechanisms allow the model to examine when shared indexical interpretations stabilize, when they fragment into multiple coexisting interpretations, and when convergence fails to occur.

4. Results

The model runs in discrete time steps. At each step, agents update their interpretive positions based on the interaction and timing rules described above. The results (see *analysis.ipynb*) indicate a consistent and structured relationship between the interpretive scale parameter (K), interaction structure, and timing regime. Across all experimental conditions, the system reliably reaches a stable configuration, with stability rates ranging from 86.7% to 100%. This suggests that the model does not produce fragile or highly contingent dynamics; instead, local interaction processes are sufficient to generate equilibrium under a wide range of parameter settings.

A first and most robust finding concerns the role of K , which systematically determines the emergent level of interpretive differentiation in the system. As K increases from 3 to 9, the mean final interpretive state increases in an approximately linear manner (from approximately 1.5 to 4.5). This indicates that K functions as a structural constraint on the interpretive space rather than a behavioral outcome.

Second, the temporal organization of interaction. While all timing regimes produce broadly similar equilibrium states, they differ substantially in the speed (steps) at which stability is reached. Asynchronous updating generally produces the fastest convergence to equilibrium, particularly under dyadic interaction. This indicates that temporal organization primarily regulates the efficiency and immediacy of local adjustment, with asynchronous updating allowing interpretive shifts to propagate without coordination delays, whereas synchronous and partially synchronous regimes introduce temporal constraints that slow system-level alignment.

At the same time, interaction structure produces a parallel but distinct effect on the same process. Dyadic interaction consistently leads to faster stabilization than group-based interaction, whereas group-based interaction consistently slows convergence, producing longer time-to-stability values across all values of K . This difference is not due to changes in the final distribution of states, which remains broadly similar across conditions, but to the fact that group interactions introduce additional local averaging over multiple neighbors. In this sense, interaction structure modulates the path dependency of the system rather than its endpoint.

Finally, the distribution of agent states at equilibrium reveals that stability does not correspond to full consensus. Instead of converging to a single interpretive value, the system stabilizes into a bounded distribution concentrated around intermediate states. Even at equilibrium, multiple interpretive positions remain present, with relatively persistent heterogeneity across the population. This suggests that stability in this model should be understood not as the final state of *enregisterment*, but as the emergence of a persistent and structured distribution of interpretive positions.

Taken together, these results indicate that stable indexical coordination can emerge without requiring uniform alignment. Rather than producing convergence to a single shared meaning, the system generates a stabilized indexical order in which differences in interpretation are both

persistent and structured. This aligns with the idea that indexical meaning is not a matter of agreement, but of socially organized differentiation: through repeated local interaction, interpretive positions become distributed across a bounded space of possible meanings, producing a layered and hierarchically constrained field of indexical value rather than a uniform consensus.

Bibliography

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